SE375 SYSTEM PROGRAMMING SPRING 2025-2026
Laboratory Assignment 12

May 22, 2026

1. Introduction

Assume that each district (Konak, Gaziemir etc) of İzmir reports their data on public transport usage and the overall data set is collected for analysis. The data set from each district contains the daily number of bus passengers and number of bus trips. An example view of a data set from one district (say Konak) is as follows:

Date	Day of the Week	Number of Trips	Number of Passengers
01.02.2023	Wednesday	34	1242
02.02.2023	Thursday	51	4589
03.02.2023	Friday	30	2568
04.02.2023	Saturday	48	3874
05.02.2023	Sunday	36	2514
06.02.2023	Monday	44	6257
07.02.2023	Tuesday	27	2674

2. Task

In this lab assignment, you are going to develop a Java Application that will calculate statistics on given data. Your application will read each data file, compute and store the values and report to the user.

2.1 Specifications

- Each data file contains the daily number of bus trips and number of bus passengers in a CSV (comma-separated values) format for each district. The first line of the file has the headers in the order of:

  Date, *Day of the Week*, *Number of Trips* and *Number of Passengers*

  and the remaining lines include data for each date of the month.
- DD.MM.YYYY is the date format.
- Day of the week is one of the below;

  Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

- The data type of *Number of Trips* and *Number of Passengers* is integer. It is guaranteed that the data files do not have missing values.
- Your application should use a single `HashMap` to store the statistics where keys are the days of the week (Monday, Tuesday etc.), and the values are:
  - The total number of passengers on that particular day of the week over all given districts,

    - The number of days corresponding to that day (e.g. "*how many Wednesdays were processed*"),
    - The average number of passengers on that particular day of the week over all given districts.
        - **IMPORTANT:** When calculating this average, don't count two rows belonging to the same date as different dates. Example: The row starting with 01.03.2023 in "Balcova.csv" and the row starting with 01.03.2023 in "Karabaglar.csv" correspond to the same date.
- Your application should **create** a thread pool with size 10.
- Your application should **assign** a thread for each data file from the thread pool and give both the file name and the HashMap as an argument. This structure should be shared between all threads.
- Each thread should **update** the HashMap while processing the given data file.
- The synchronization should be achieved using Locks
- Do **NOT** display the data in the HashMap yet. The main application should not update the HashMap at any point.

SECURE THE RESULTS

- Once threads are finished, your application now should store the results in the Hash Map and the message digest into a file **encrypted**. You will store the values that you have calculated before. Namely:
    i. The total number of passengers on that particular day of the week over all given districts,
    ii. The number of days corresponding to that day (e.g. "*how many Wednesdays were processed*"),
    iii. The average number of passengers on that particular day of the week over all given districts.
- The encryption should be symmetric. You must generate the key for that.
- Calculate the message digest of the file using SHA-256.
- Write the message digest as the last entry in the file and encrypt the whole file.
- Read the contents of the file. Decrypt and display the results.

2.2 Example Runs

The result should look like the following:

```
Parsing Karsiyaka.csv...
Parsing Konak.csv...
Parsing Seferihisar.csv...
Parsing Bornova.csv...
Parsing Gaziemir.csv...
Parsing Karabaglar.csv...
Parsing Balcova.csv...
Parsing Urla.csv...
Threads are complete. Here are the statistics for the number of passengers:

Day of the week      Total Passengers     Number of days       Avg Passengers
--------------       ----------------     --------------       --------------
Tuesday              78109                4                    19527.25
Friday               96800                5                    19360.0
Thursday             92935                5                    18587.0
Sunday               78475                4                    19618.75
Saturday             80456                4                    20114.0
Monday               80998                4                    20249.5
Wednesday            97133                5                    19426.6
```

3 Regulations

- You should use Java.
- Late submission is **not** allowed.
- This is an **individual** assignment. Plagiarism, copying, cheating, outsourcing the exam to another person or organization for with or without pay are considered as actions of academic dishonesty. Failure to maintain academic honesty may result in disciplinary action according to the Izmir University of Economics' disciplinary bylaw for students of institutions of higher education (https://www.ieu.edu.tr/en/bylaws/type/read/id/13).
- Submissions should be done to Blackboard. Submissions via e-mail will not be accepted.
- You should make sure to obey the specifications.

4 Submission

- You should submit 2 java files to Blackboard containing your Java files.
- By submitting you pledge the following; *"On my honor, as an Izmir University of Economics student, I affirm that I have not given or received any unauthorized help on this exam, and that this work is my own."*